

$$= 1.3 \times 10^{-20} \text{ N m}$$

- 20 D** An α -particle has mass $4u$ and carries charge $+2e$.

Within the plates, constant $E = V/x$

By PCOE,

Loss in KE = Gain EPE

$$\frac{1}{2} (4u)(v_i^2 - v_f^2) = qV$$

$$\frac{1}{2} (4u)(v_i^2 - v_f^2) = 2eEx$$

$$v_f^2 = v^2 - eEx/u$$

$$v_f = \sqrt{v^2 - eEx/u}$$

- 21 D** $R = R_1 + R_2 = \frac{\rho 2x}{t} + \frac{\rho x}{t}$ where ρ is resistivity and t is thickness